

Quantifying the Effects of Numerical Diffusion During Galaxy Formation in RAMSES

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Abstract

Numerical simulations are a powerful tool used to study galaxy formation and evolution, but their accuracy is limited by numerical errors inherent to computational methods. In this work, we investigate the impact of numerical diffusion and numerical dispersion in the computational astrophysics code RAMSES, with a focus on how these errors influence the evolution of self-gravitating systems. We first demonstrate the presence of velocity-dependent numerical diffusion in a galaxy-scale simulation using AGORA initial conditions, where regions with large absolute velocities relative to the computational grid exhibit artificially smoothed structure. To isolate and quantify these effects, we perform idealized simulations of sinusoidal density perturbations evolving in periodic, moving boxes under hydrodynamics and self-gravity. By tracking the evolution of the root mean squared (RMS) density, we show that numerical diffusion coefficients take the form $D \propto v \Delta x^2 k$. We quantify numerical dispersion by measuring deviations in the angular frequency of stable perturbations, discovering that phase errors also change with velocity and resolution. This work highlights the importance of quantifying and mitigating numerical errors to ensure that simulation results are governed by physical processes rather than discretization effects.

Keywords: diffusion – dispersion – hydrodynamics – methods: numerical – galaxies: evolution

1 Introduction

Galaxies are massive, self-gravitating systems composed of stars, gas, dust, and dark matter, spanning tens to hundreds of kiloparsecs in size and evolving over cosmic timescales. The baryonic component of galaxies contain cold, dense regions of gas and dust forming giant molecular clouds (GMCs), which typically contain more than $10^5 M_\odot$ of material and serve as the primary locations of star formation [Solomon et al. \(1987\)](#). Within these clouds, star formation is triggered when local density overcomes internal gas pressure, resulting in a gravitationally unstable cloud that collapses.

Direct observations of galaxies provide valuable insight into their structure and composition, but are inherently limited to snapshots at a specific time. As a result, simulations have become an essential tool used by astrophysicists to study large-timescale galaxy evolution, and en-

able controlled testing of physical models. A variety of numerical methods have been developed and can be broadly categorized into particle-based and grid-based schemes. Particle-based methods, such as Smoothed Particle Hydrodynamics (SPH), represent fluids using discrete elements that naturally follow the mass distribution. In contrast, grid-based methods solve the equations of fluid dynamics on a computational mesh, offering improved accuracy in resolving shocks and discontinuities. A widely used grid-based code is *RAMSES*, an Adaptive Mesh Refinement (AMR) code built in Fortran 90 [Teyssier \(2002\)](#). The AMR framework allows the grid to be dynamically refined in regions of interest, enabling high-resolution simulations while maintaining computational efficiency. RAMSES solves the Euler equations of hydrodynamics in conservative form using a second-order Godunov scheme with piecewise linear reconstruction, and includes self-gravity by solving Poisson's equation for the gravitational potential, which is

coupled to the fluid evolution as a source term.

Despite their capabilities, galaxy simulations face significant challenges due to the enormous dynamic range involved. Structures in astronomy span many orders of magnitude, from dark matter halos (~ 100 kpc) to molecular clouds (~ 100 pc) and down to star-forming cores ($\ll 1$ pc) [Krumholz et al. \(2019\)](#). Resolving all relevant scales simultaneously remains computationally infeasible, necessitating the use of subgrid models to handle unresolved processes such as star formation and feedback. Consequently, simulations must balance resolution and computational cost, while minimizing numerical artifacts that can influence physical results over time. One such artifact is numerical diffusion – a purely numerical smoothing effect that artificially smooths structure as fluid elements move across the computational grid. Unlike physical diffusion, numerical diffusion depends on resolution and the velocity of the flow relative to the grid, thereby breaking Galilean invariance. As a result, the evolution of structures in a simulation can depend on their absolute velocity rather than purely on relative motion. In the context of galaxies, which undergo both bulk translational motion and internal rotation, this leads to spatially varying numerical effects. Regions where velocities align with each other experience enhanced diffusion, altering the structure of GMCs and shifting their stability properties.

Star formation in these environments is commonly understood through the framework of Jeans instability, which describes the competition between pressure support and self-gravity [Jeans \(1902\)](#). The stability is determined by the dispersion relation,

$$\omega^2 = c_s^2 k^2 - 4\pi G \rho_0, \quad (1)$$

where c_s is the sound speed, k is the perturbation wavenumber, G is the gravitational constant, and ρ_0 is the gas density. The dispersion relation is derived from the Euler continuity and momentum fluid equations, and the Poisson equation for gravity. If we define $\vec{\rho}(\vec{x}, t) = A e^{\vec{k} \cdot \vec{x} - \omega t}$ to be a density perturbation then, the stability of the perturbation is dictated by the sign of ω^2 . When the pressure support term is greater than the gravity term, $\omega^2 > 0$ and the solutions are sinusoidal which leads to oscillations. When the gravity term dominates, $\omega^2 < 0$ and the solutions are exponentials. One mode of the perturbation decays as $-\omega$, and the other mode grows as $+\omega$, representative of a gravitational collapse. This behaviour can be seen in Figure 1. Another way to charac-

terize this behaviour is if a perturbation of wavelength λ exceeds the Jeans length,

$$\lambda_J = \sqrt{\frac{\pi c_s^2}{G \rho}}, \quad (2)$$

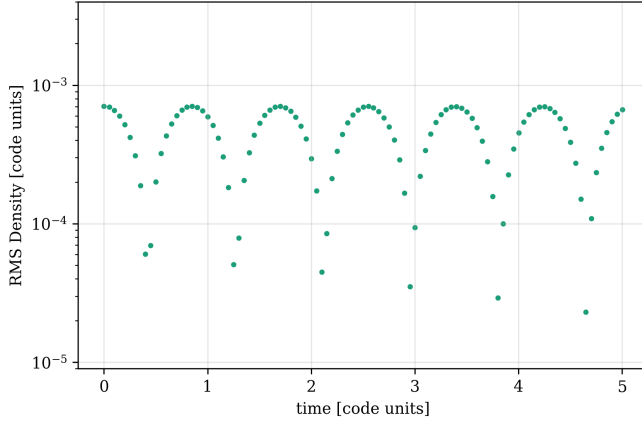
then a gravitational collapse occurs, while smaller perturbations oscillate like a sound wave. When numerical diffusion terms of the form $(D_v \nabla^2 \vec{v}, D_\rho \nabla^2 \vec{\rho})$ are introduced into the Euler fluid equations, the resulting Jeans dispersion relation is modified to

$$\omega^2 = c_s^2 k^2 - 4\pi G \rho_0 + D_\rho D_v k^4 - i(D_\rho + D_v) k^2 \omega. \quad (3)$$

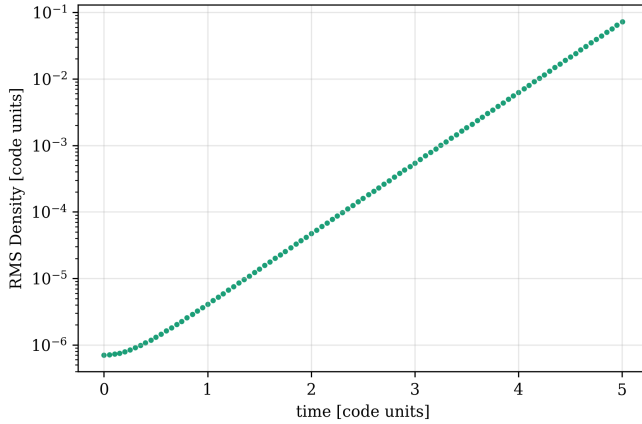
The additional real and imaginary terms arise purely from discretization and modify wave dynamics. In particular, the imaginary component introduces exponential decay of perturbation, while the higher-order k^4 term alters the stability threshold. Similarly, numerical dispersion, associated with ∇^3 derivatives, modifies the angular velocity ω , introducing errors in their propagation speeds. Together, these numerical errors alter both the growth rates and oscillatory behaviour of perturbations, leading to deviations from the classical Jeans instability.

The goal of this work is to quantify the impact of numerical diffusion in galaxy simulations performed with RAMSES, and to assess whether these errors can lead to measurable deviations in the evolution of self-gravitating structures. In particular, we investigate how flow velocity influences the stability and evolution of density perturbations.

This report is organized as follows. In Section 2, we describe the simulation setup, including modifications to RAMSES and the initial conditions. Section 3 presents the results of our numerical experiments, focusing on diffusion and dispersion effects. In Section 4, we discuss the implications of our findings and outline directions for future work.



(a) $\omega^2 = 13.75$ (Stable Sound Wave)



(b) $\omega^2 = -5.89$ (Gravitational Collapse)

Figure 1: Root mean squared (RMS) density versus time plot of a perturbation $\rho(x) = 10^{-6} \sin(2\pi x)$ evolving under gravity and hydrodynamics in an $L = 1$ (code units) periodic stationary box. Simulation run for $t \in [0, 5]$ (code units) with outputs recorded at intervals of $\Delta t = 0.05$.

2 Methods

2.1 Simulation Workflow

All simulations in this work were performed by modifying RAMSES to create custom perturbations for systems evolving under self-gravity and hydrodynamics. Simulations were executed on Rorqual high-performance computing (HPC) cluster provided by the [Digital Research Alliance of Canada \(2023\)](#). Post-processing and analysis were carried out in Python using the yt analysis toolkit [Turk et al. \(2011\)](#), which naively supports RAMSES outputs for visualization and quantitative analysis. Metrics such as root mean squared (RMS) density, velocity fields, and wave properties were computed directly from the

simulation outputs in code units.

2.2 Galaxy-Scale Demonstration

To demonstrate the impact of numerical diffusion in a realistic galaxy simulation, we performed simulations of an isolated disk galaxy using initial conditions based on the AGORA project [Kim & et al. \(2014\)](#). The galaxy was initialized with counterclockwise rotation velocity of 200 rad/s and a uniform translation velocity of 200 km/s in the positive x-direction. The purpose of this setup is to create a velocity gradient across the galaxy. In regions where the rotational and translational velocity vectors are aligned, the magnitude of the total velocity is maximized, whereas the regions where they are anti-parallel, the velocity is minimized. Since numerical diffusion depends on the absolute velocity of the flow relative to the grid, this configuration provides a direct test of whether galactic structures, like spiral arms, are smoothed in high-velocity regions in practice.

2.3 Idealized Perturbation Setup

To isolate the effects of numerical errors from the complexity of galaxy-scale simulations, we construct a simplified model consisting of one-dimensional density perturbations in uniform, periodic boxes evolving under self-gravity and hydrodynamics. The initial density field is defined as

$$\rho(x) = \rho_0 + A \sin\left(\frac{2\pi x}{\lambda} - \phi\right), \quad (4)$$

where A is the perturbation amplitude, λ is the wavelength, and ϕ is the arbitrary phase shift. The gas is modeled using an adiabatic equation of state,

$$P = P_0 \left(\frac{\rho}{\rho_0}\right)^\gamma, \quad (5)$$

with the adiabatic index $\gamma = \frac{5}{3}$. A uniform translational velocity v_0 is imposed in the positive x-direction to replicate different regions of a moving galaxy. This controlled setup enables direct comparisons with linear perturbation theory and analytic expectations from Jeans instability.

2.4 Quantifying Numerical Diffusion

Numerical diffusion was quantified by tracking the decay and growth of perturbation over time. We measured the

evolution of the RMS density,

$$\rho_{\text{RMS}}(t) = \sqrt{\langle (\rho - \rho_0)^2 \rangle}, \quad (6)$$

as a function of time. Two sets of experiments were conducted:

(i) Resolution Dependence The density perturbation given by Equation 4 with $\phi = 0$ rad, an amplitude $A = 10^{-6}$ and wavelength $\lambda = 1$ was initialized in a periodic domain. The pressure was perturbed with $P_0 = 0.4$ for a Jeans stable sound wave. The system was evolved over the time interval $t \in [0, 10]$ (code units), with outputs recorded at intervals of $\Delta t = 0.05$. Simulations were performed across a range of velocities,

$$v_0 = [0, 1, 2, 4, 8, 16, 32, 64], \quad (7)$$

and spatial resolutions $\Delta x = [2^{-5}, 2^{-6}, 2^{-7}, 2^{-8}]$. This test allows us to analyze how numerical diffusion scales with resolution.

(ii) Scale Dependence We repeated the above experiment at a fixed resolution $\Delta x = 2^{-8}$ while varying the perturbation wavenumber, $k = [2\pi, 4\pi, 6\pi, 8\pi]$, for the velocities,

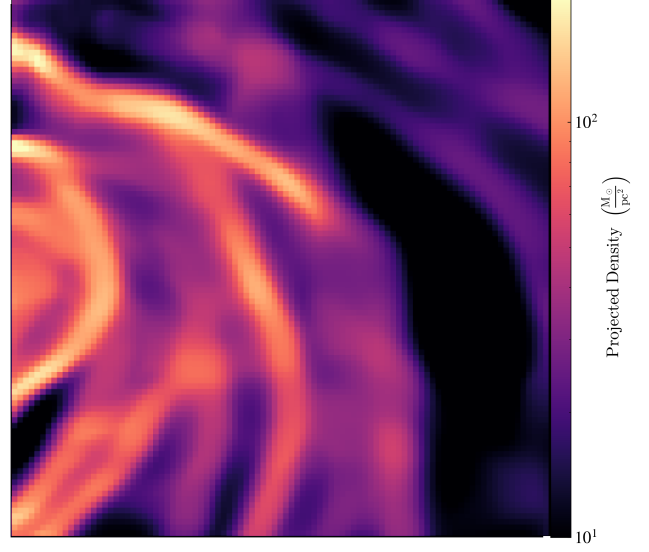
$$v_0 = [0, 1, 2, 4, 8, 16]. \quad (8)$$

This test was used to measure how the effects of numerical diffusion change with the scale of a structure

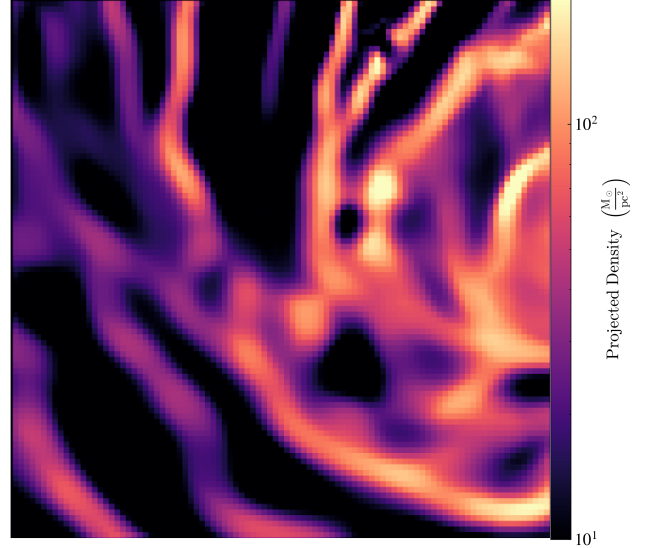
2.5 Quantifying Numerical Dispersion

In addition to density evolution, numerical dispersion was quantified by measuring changes in the angular frequency over time. The numerical frequency ω_{num} was extracted from the temporal evolution of the RMS density field and compared to the analytical value ω_{exact} from Jeans instability. To isolate dispersion effects, these simulations were performed without self-gravity, ensuring that the perturbations remained stable sound waves. The setup follows that Section 2.4, with simulations conducted for the velocities in Equation 7, and spatial resolutions $\Delta x = [2^{-6}, 2^{-7}, 2^{-8}]$.

3 Results



(a) Region with a high absolute velocity, where the translational and rotational velocities are aligned.



(b) Region with low absolute velocity, where the translational and rotational velocities cancel.

Figure 2: Projected surface density of the AGORA initial-condition galaxy after 108 Myr. The two panels compare regions with different absolute velocities relative to the computational grid. The high-velocity region shows visibly smoother structure, while the lower-velocity region retains sharper spiral features. This contrast illustrates the velocity-dependence of numerical diffusion in grid-based simulations.

Figure 2 shows snapshots of the projected surface density maps from the galaxy-scale demonstration after 108 Myr of evolution. The region with the higher absolute velocity exhibits noticeably more smearing of the spiral structure compared to the low-velocity region that retains sharper density gradients. This behaviour is consistent with numerical diffusion acting more strongly when fluid structures move quickly across the grid. In a Galilean-invariant system, the physics of star formation should not be influenced by the velocity of the reference frame. However, the results of this test provide a demonstration of velocity-dependent smoothing. This galaxy-scale test serves a motivating example showing that numerical diffusion is not simply a theoretical concern, but an error that alters resolved structures in a realistic galaxy simulation, even before any simplified idealized tests are considered.

The idealized perturbation experiments were designed to isolate numerical diffusion from the complexity of galaxy-scale structure. Figure 3 summarizes the measured RMS density decay rates from the diffusion tests. For the resolution, the perturbation amplitude decays as velocity v_0 increases, indicating that numerical damping becomes stronger at higher flow speeds relative to the grid. At a fixed velocity, the decay also becomes stronger at coarser resolutions, consistent with the expectation that numerical diffusion decreases as the cell size decreases. The wavenumber study shows an additional scale dependence. Shorter wavelength perturbations experience larger amounts of damping compared to longer wavelength perturbations, indicating that the small-scale structure is more susceptible to numerical smoothing. This is expected because high-wavenumber modes are less resolved by a fixed grid and therefore suffer larger truncation errors. Combining the results show that the rate at which numerical diffusion suppresses amplitude depends on both the absolute flow velocity and wavenumber of the perturbation. The form that the diffusion coefficients take in RAMSES is

$$D \propto v \Delta x^2 k \quad (9)$$

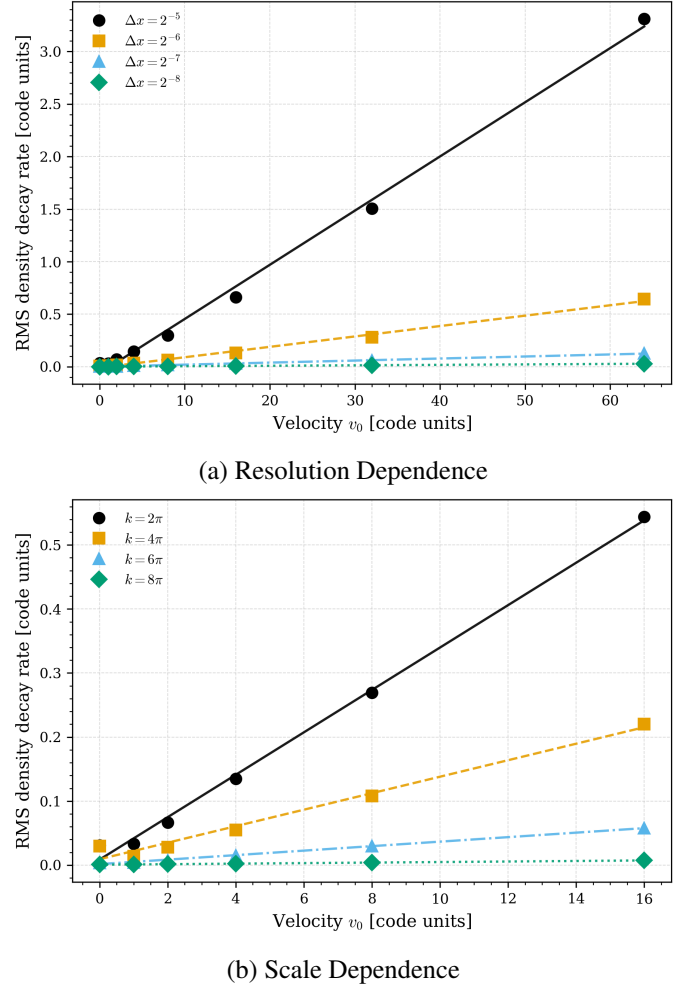


Figure 3: Measured decay rate of the perturbation RMS density as a function of velocity v_0 . The resolution dependence panel varies resolution at fixed wavelength $k = 2\pi$, while the scale dependence panel varies wavenumber at fixed resolution $\Delta x = 10^{-8}$. In both cases, steeper slopes indicate stronger numerical damping of the perturbation amplitude.

Figure 4 shows the measured angular frequency of the perturbation as a function of velocity and resolution. The numerical frequency, ω_{num} , remains close to the expected value of $|\omega| \approx 10.26$ at low velocities and high resolution, but increasingly deviates as the velocity increases and the grid becomes coarser. This indicates that the phase propagation of the wave is also affected by discretization errors. In physical terms, numerical dispersion modifies the speed at which waves propagate through the domain. For an oscillating perturbation, this can introduce phase shifts that accumulate over time. The result is a mode that may deviate from the analytic expectation. Although the dispersion errors are smaller than the amplitude damping in some of the runs, they remain

significant because they alter the frequency of oscillations and can change the stability of a perturbation. This becomes especially relevant in long simulations, where small phase errors can accumulate into noticeable differences in the evolution of density structure.

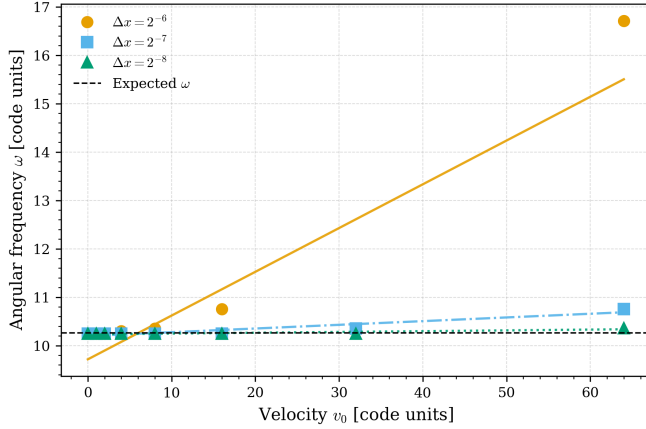


Figure 4: Measured angular frequency ω of the perturbation as a function of bulk velocity and spatial resolution. The numerical frequency was obtained from the time evolution of the oscillation phase and compared with the analytical frequency $|\omega| \approx 10.26$ expected from the initial conditions. Deviations from the expected value increase with flow speed and decrease with higher resolution, indicating the presence of numerical dispersion.

4 Discussion

The results of this work demonstrate that numerical diffusion and numerical dispersion are both present in RAMSES and that their effects depend on the motion of the fluid relative to the computational mesh, the spatial resolution, and the scale of the perturbation. The galaxy-scale tests provide direct visual evidence that structures in high-velocity regions experience more smearing than those in low-velocity regions, indicating a breakdown of Galilean invariance at the numerical level.

The idealized perturbation experiments confirm this behaviour in a controlled system. The decay of density perturbations increases with velocity, decreases with resolution, and becomes stronger at smaller spatial scales. In addition, numerical dispersion introduces phase errors that alter the propagation speed of waves. Together, these effects modify the evolution of perturbations, and can influence whether a structure remains stable or undergoes gravitational collapse.

These results have important implications for grid-based galaxy simulations. Under the framework of Jeans instability, star formation is sensitive to the growth of density perturbations. Velocity-dependent numerical diffusion can introduce purely numerical variations in collapse behaviour. As a result, simulation outcomes may not only depend on the physical model, but also on the numerical artifacts that arise from these errors.

Several extensions of this work can further improve our understanding of numerical effects in galaxy simulations. First, more simulations could be performed to explore different regions of the dispersion relation, including numerical error-driven transitions between stable and unstable regimes. This would allow for a more complete characterization of how numerical errors affect star formation. Second, more realistic perturbations should be considered. The sinusoidal perturbations used in this work provide simple, controlled tests, but real astrophysical systems, such as molecular clouds, exhibit complex, turbulent structure. Studying more physically realistic density fields would help determine how numerical diffusion operates in practical simulations. Finally, it would be valuable to develop quantitative diagnostics or correction schemes to measure and mitigate numerical diffusion in simulations.

5 Conclusion

In this work, we investigated the impact of numerical diffusion and dispersion in galaxy simulations performed with RAMSES. We first demonstrated the presence of velocity-dependent numerical smoothing in a realistic galaxy simulation using the AGORA initial conditions. Regions with higher absolute velocities relative to the computational grid contained visibly smoother structure, indicating that numerical diffusion can alter galaxy evolution. The idealized perturbation tests isolated the numerical errors and demonstrated that both are present and measurable within RAMSES. For numerical diffusion, the form of the diffusion coefficients was determined. Overall, this study shows that numerical diffusion and dispersion can significantly influence the evolution of density structures in simulations, particularly in systems with large absolute velocities. These effects can modify both the structure and stability of perturbations, and therefore must be considered when interpreting results from galaxy simulations.

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